

One Shock, Five Trajectories: Comparing Real GDP Growth and Consumer-Price Inflation Across Five Economies (2019–2023)

Data Lab Research Note I — Descriptive Observations from the World Bank Open Data Series

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Introduction — Why I Built a Cross-Country Comparison

When an economy experiences a sharp rise in inflation or a sudden swing in output growth, the instinctive tendency in public debate is to treat the episode as entirely domestic — or, conversely, to blame it entirely on a global shock beyond any government’s control. Living through Turkey’s post-2020 inflation surge while reading my first textbooks in macroeconomics made me sceptical of both extremes. If the pandemic and the subsequent energy price shock were global phenomena, why did the inflation trajectories that followed look so different across countries? Conversely, if domestic policy was the sole determinant, why did nearly every major economy experience an inflection point at almost the exact same moment between 2021 and 2022?

I built the Macroeconomic Data Lab to test my own intuition against harmonised public data. Rather than relying on isolated headlines or incompatible national definitions, the application pulls two standardized series directly from the World Bank Open Data API — annual real GDP growth (NY.GDP.MKTP.KD.ZG) and annual consumer-price inflation (FP.CPI.TOTL.ZG) — across five economies between 2019 and 2023: Turkey, the United States, Germany, the United Kingdom, and Japan. As a high school student learning empirical economics, my goal in this first research note is not to claim causal proof, but to document what a side-by-side comparison reveals — and where a two-variable lens reaches its limits.

Section I — The Shared Contraction and the Asymmetric Rebound (2019–2021)

Looking at the real GDP growth series in the Data Lab, the first pattern that stands out is the contrast between the uniformity of the 2020 shock and the asymmetry of the 2021 rebound. In 2020, four of the five economies in the sample contracted sharply: output fell by 2.2 percent in the United States, 4.1 percent in both Germany and Japan, and 10.0 percent in the United Kingdom. Turkey was the sole exception in the cohort, recording positive real growth of 1.9 percent in 2020 before surging to 11.4 percent in 2021 — the highest single-year growth rate in the entire five-country dataset.

Economy (World Bank Series)	2019	2020	2021	2022	2023	5Y Mean
Turkey — Real GDP Growth (%)	0.9	1.9	11.4	5.5	5.1	5.0%
United States — Real GDP Growth (%)	2.3	-2.2	6.2	2.5	2.9	2.3%
Germany — Real GDP Growth (%)	1.1	-4.1	3.7	1.4	-0.2	0.4%

United Kingdom — Real GDP Growth (%)	1.6	-10.0	8.5	5.1	0.3	1.1%
Japan — Real GDP Growth (%)	-0.4	-4.1	2.6	1.0	1.4	0.1%
Turkey — CPI Inflation (%)	15.2	12.3	19.6	72.3	53.9	34.7%
United States — CPI Inflation (%)	1.8	1.2	4.7	8.0	4.1	4.0%
Germany — CPI Inflation (%)	1.4	0.5	3.1	6.9	5.9	3.6%
United Kingdom — CPI Inflation (%)	1.8	0.9	2.6	9.1	7.3	4.3%
Japan — CPI Inflation (%)	0.5	0.0	-0.2	2.5	3.3	1.2%

Table 1: Annual Real GDP Growth and Consumer-Price Inflation (2019–2023). Source: World Bank Open Data API v2.

Read in isolation, Turkey’s 2020–2021 growth figures might look like an unambiguous policy success. Yet placing the GDP series directly beside the CPI inflation series in the Data Lab immediately complicates that reading. Turkey’s rapid credit-fuelled expansion during the pandemic — followed by the unorthodox monetary easing of late 2021 that I examined in my essay on the Central Bank’s credibility gap — meant that when global supply bottlenecks and energy shocks arrived in 2022, domestic inflation expectations were already unanchored.

Section II — The Inflation Divergence of 2022–2023

Between 2021 and 2022, consumer-price inflation rose in every economy in the sample. This shared upward movement confirms that a common external shock — post-lockdown supply constraints and the 2022 energy price spike — was genuinely at work. In the United States, inflation rose from 4.7 percent to 8.0 percent before moderating to 4.1 percent in 2023 as the Federal Reserve tightened policy. In Germany and the United Kingdom, which were more directly exposed to European natural gas disruptions, inflation peaked at 6.9 percent and 9.1 percent in 2022 and remained elevated in 2023 while output growth stalled (-0.2 percent in Germany and 0.3 percent in the UK). Even Japan, after decades of near-zero inflation, saw CPI reach 3.3 percent by 2023.

Yet Turkey’s path belongs to a different order of magnitude altogether. Annual CPI inflation jumped from 19.6 percent in 2021 to 72.3 percent in 2022 and 53.9 percent in 2023 — more than eight times the five-year average of any advanced economy in the cohort. What the Data Lab makes visually unmistakable is that a global shock does not simply add a fixed number of percentage points to every country’s inflation rate; it interacts with the domestic institutional framework. Where central bank credibility anchored long-run expectations, the 2022 price surge behaved like a temporary shock, albeit a painful one. Where monetary policy cut interest rates into the teeth of rising inflation, the global shock triggered exchange-rate pass-through and self-reinforcing price-setting behaviour.

Section III — What This Comparison Establishes, and What It Cannot

Working with the World Bank API taught me a discipline that is easy to overlook when writing purely qualitative essays: descriptive data can rule out bad explanations, even when it cannot prove a single causal one. Comparing these five series side by side shows that Turkey's inflation crisis cannot be dismissed as a mere reflection of global post-pandemic inflation, just as Germany's 2023 stagnation cannot be understood without looking at the trade-off between energy exposure and disinflation. At the same time, two annual series for five countries — twenty-five observations per indicator — cannot isolate how much of each country's path came from interest rates, fiscal deficits, import composition, or wage-setting institutions. Recognizing that boundary is the whole point of the Data Lab: it is a starting point for asking sharper empirical questions, not a substitute for estimating a full structural model.
